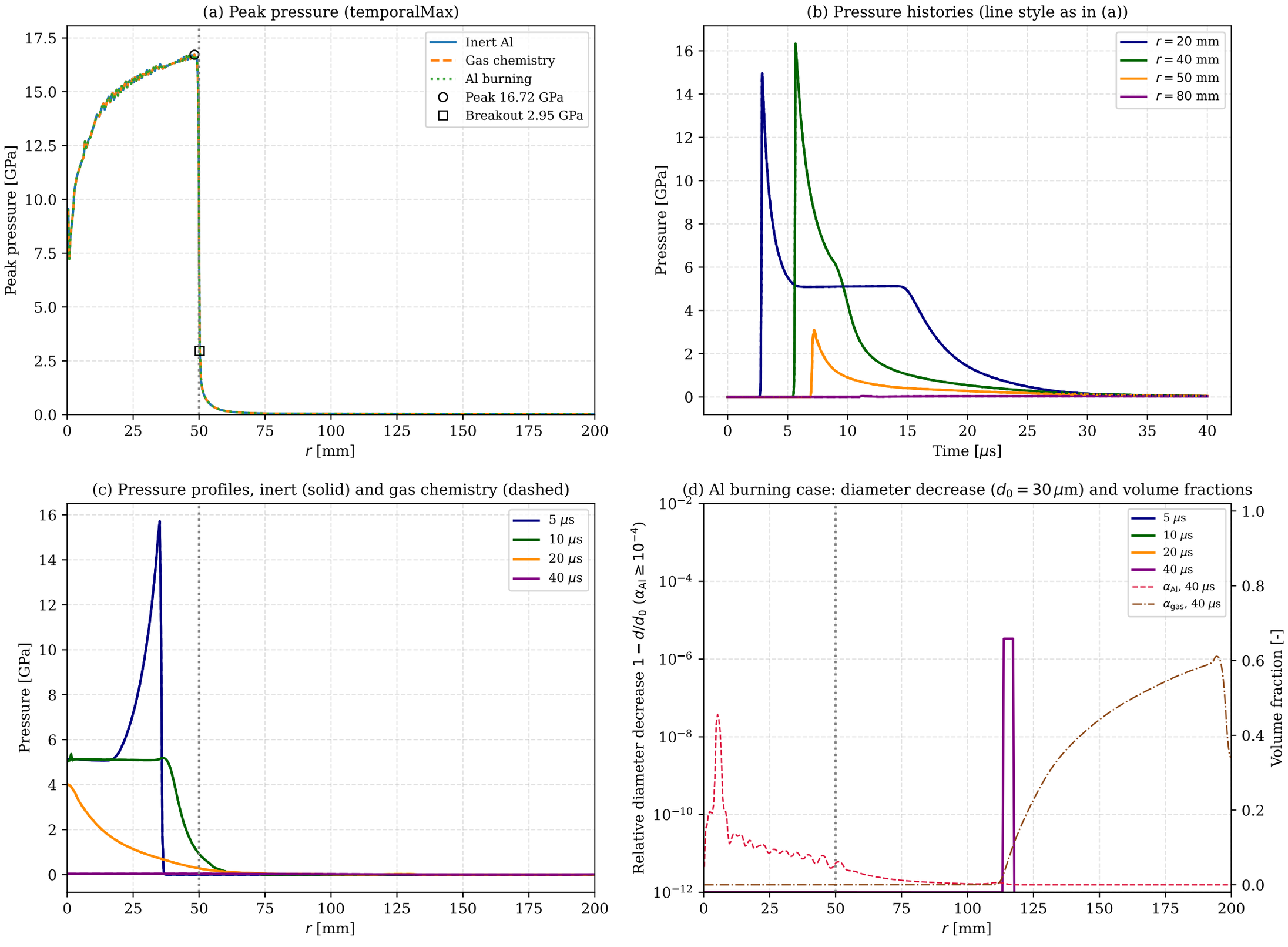
(a) Peak pressure (temporalMax)
Peak pressure [GPa]
r [mm]
Inert Al
Gas chemistry
Al burning
Peak 16.72 GPa
Breakout 2.95 GPa
(b) Pressure histories (line style as in (a))
Pressure [GPa]
Time [μs]
r = 20 mm
r = 40 mm
r = 50 mm
r = 80 mm
(c) Pressure profiles, inert (solid) and gas chemistry (dashed)
Pressure [GPa]
r [mm]
5 μs
10 μs
20 μs
40 μs
(d) Al burning case: diameter decrease (d0 = 30 μm) and volume fractions
Relative diameter decrease 1 − d/d0 (αAl ≥ 10−4)
Volume fraction [-]
r [mm]
5 μs
10 μs
20 μs
40 μs
αAl, 40 μs
αgas, 40 μs